TEACHAI · PEDAGOGICAL LESSON PLAN

Mastering Chemical Reactions and Equations

APPROVED

Subject: Science	**Grade:** Grade 10	**Duration:** 45 mins
Topic: Chemical Reactions and Equations	**Curriculum:** CBSE / Computer Science	**Teacher:** Authorized Teacher

LEARNING OBJECTIVES

- Identify different types of chemical reactions (Combination, Decomposition, Displacement, Double Displacement, Oxidation-Reduction)
- Write and balance chemical equations correctly using the law of conservation of mass

PREREQUISITES & PRIOR KNOWLEDGE

- Basic understanding of atoms, molecules, and chemical formulas
- Law of conservation of mass

INTRODUCTION / WARM-UP (5 MINS)

Hook: Show a quick video clip of a burning magnesium ribbon or rusting iron. Ask students: 'What changed and how do we represent this scientifically?'

Prior Knowledge Check: Briefly review what reactants and products are and how to write basic chemical formulas.

MAIN CONCEPTS & CORE INSTRUCTION

Balanced Chemical Equations: A chemical equation represents a chemical reaction using symbols and formulas. It must be balanced to obey the Law of Conservation of Mass.

Key Vocabulary: Reactant, Product, Coefficient, Balanced Equation

Types of Chemical Reactions: Reactions are classified into categories based on how atoms rearrange: Combination, Decomposition, Displacement, Double Displacement, and Redox.

Key Vocabulary: Combination, Decomposition, Displacement, Oxidation, Reduction

STEP-BY-STEP TEACHING ACTIVITIES & PACING

Phase	Time	Teacher Action	Student Action
Introduction & Hook	5m	Facilitate the hook demonstration and elicit prior knowledge on chemical changes.	Observe the demonstration and share initial thoughts on what is happening at the molecular level.
Direct Instruction	15m	Explain the step-by-step process of balancing chemical equations and introduce the 5 main types of reactions with examples.	Take guided notes and participate in mini-check-ins by raising hands to identify reaction types.
Guided Practice	15m	Work through a complex balancing problem on the whiteboard and guide students through classifying reactions.	Work in pairs to balance a given unbalanced equation on mini whiteboards.

DIFFERENTIATED STUDENT ACTIVITIES

Reaction Type & Balancing Lab Station (Pair): Examine 4 reaction cards, classify their reaction type, and balance the equations provided on the worksheet.

- **Remedial:** Provide pre-counted atom inventory tables to help balance equations.
- **Standard:** Complete standard worksheet classifying and balancing 4 unique equations.
- **Advanced:** Create two original chemical equations of different types and challenge a partner to balance them.

ASSESSMENT & EVALUATION

- **Exit Ticket:** What type of reaction is represented by: $CaCO_3 \rightarrow CaO + CO_2$?
  Expected Answer: Decomposition reaction
- **Exit Ticket:** Why must chemical equations be balanced?
  Expected Answer: To obey the Law of Conservation of Mass.

HOMEWORK & EXTENSION ACTIVITIES

Assignment: Complete textbook exercises on balancing equations (Questions 1 to 5).

Advanced Challenge: Research a real-world industrial application of a redox reaction and write a 1-page summary.

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